

Research Paper

Title:

Instructor Performance and Course Quality Evaluation on EduPro using Python and Streamlit

Abstract

This project presents an interactive dashboard for evaluating instructor performance and course quality on the EduPro online learning platform. The dashboard is developed using Python, Streamlit, Pandas, and Plotly. It analyzes teacher ratings, course ratings, transaction details, and user information from multiple Excel sheets. The system merges the data and presents it in an easy-to-understand visual format. Interactive filters allow users to analyze different expertise areas, course categories, and course levels. The dashboard helps administrators identify top-performing instructors, monitor course quality, and make better business decisions.

Keywords: Python, Streamlit, Dashboard, Data Analysis, Plotly, Pandas, Online Learning

1. Introduction

Online learning platforms generate a large amount of educational data every day. Administrators need efficient tools to evaluate instructor performance and course quality.

The objective of this project is to create a dashboard that provides valuable insights from the EduPro platform.

The dashboard displays:

- Teacher performance
- Course ratings
- User statistics
- Transaction details
- Interactive charts
- Teacher leaderboard

This helps administrators make data-driven decisions.

2. Problem Statement

Managing educational data manually is difficult because the information is stored across multiple tables.

Without visualization,

- Performance comparison becomes difficult.
- Decision making becomes slow.
- Important trends remain hidden.

Therefore, an interactive dashboard is required.

3. Objectives

The main objectives are:

- Analyze instructor performance.
- Analyze course quality.
- Display important KPIs.
- Create interactive visualizations.
- Generate teacher rankings.
- Allow filtering of data.
- Improve decision making.

4. Tools and Technologies

Tool	Purpose
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Python	Programming Language
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Streamlit	Dashboard Development
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Pandas	Data Processing
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Plotly	Interactive Charts
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VS Code	Code Editor
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Excel	Dataset
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GitHub	Version Control
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5. Dataset Description

The project uses an Excel workbook containing four sheets.

Teachers

Contains

- Teacher ID
- Teacher Name
- Age
- Gender
- Expertise
- Experience
- Rating

Courses

Contains

- Course ID
- Course Name
- Category
- Level
- Course Rating

Users

Contains

- User ID
- User Name

Transactions

Contains

- Transaction ID
- Teacher ID
- User ID

- Course ID
 - Date
 - Payment Method
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6. Methodology

The project follows these steps:

Step 1

Load Excel sheets.



Step 2

Merge datasets.



Step 3

Clean the data.



Step 4

Create KPIs.



Step 5

Apply filters.



Step 6

Generate charts.



Step 7

Display dashboard.

7. Dashboard Features

The dashboard includes:

KPI Cards

- Total Teachers
 - Total Courses
 - Total Users
 - Total Transactions
 - Average Teacher Rating
 - Average Course Rating
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Filters

- Expertise
 - Course Category
 - Course Level
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Visualizations

- Teacher Rating Distribution
 - Average Course Rating by Category
 - Teacher Leaderboard
 - Dataset Preview
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Download Button

Users can download filtered data as CSV.

8. Results

The dashboard successfully displays

- Interactive metrics
- Teacher rankings
- Course analysis

- Category comparison
- Dynamic filtering
- Download functionality

The dashboard provides quick insights into instructor performance.

9. Advantages

- Easy to use
 - Interactive
 - Fast analysis
 - Attractive interface
 - No SQL required
 - Simple deployment
 - Helps management decisions
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10. Limitations

- Uses static Excel data.
 - No real-time database.
 - No login system.
 - No prediction model.
 - Limited advanced analytics.
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11. Future Scope

The project can be improved by adding

- Machine Learning prediction
- Student feedback analysis
- Database integration
- Cloud deployment
- User authentication

- Email reports
 - AI recommendations
 - Mobile responsive design
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12. Conclusion

The EduPro Dashboard successfully analyzes instructor performance and course quality using Python and Streamlit. It transforms raw Excel data into meaningful visual insights through charts, KPIs, and interactive filters. The dashboard helps educational administrators monitor instructor performance, evaluate courses, and make informed decisions efficiently. It demonstrates how modern data visualization tools can improve educational analytics.

References

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